PC Serial Port Control

This article demonstrates how to control the servo angles, buzzer, RGB lights, and read servo angles of the miniArm using the PC serial.

1. Working Principle

1)By connecting miniArm to TTL serial and interfacing it with a PC, serial communication is established to control miniArm through the serial port.

2)Communication Protocol:

There are some examples of communication protocols. The protocol instruction format is as follows:

Frame Header	Function code	Data length	Data information	Parity
0xAA 0x66	func	len	data	check

Header: When the header is received, it indicates the presence of data transmission, and the header data length is fixed at 2 bytes;

Function Code: It indicates the function of an information frame;

Data Length: It indicates the amount of data that follows;

Data Information: It indicates the transmitted data information;

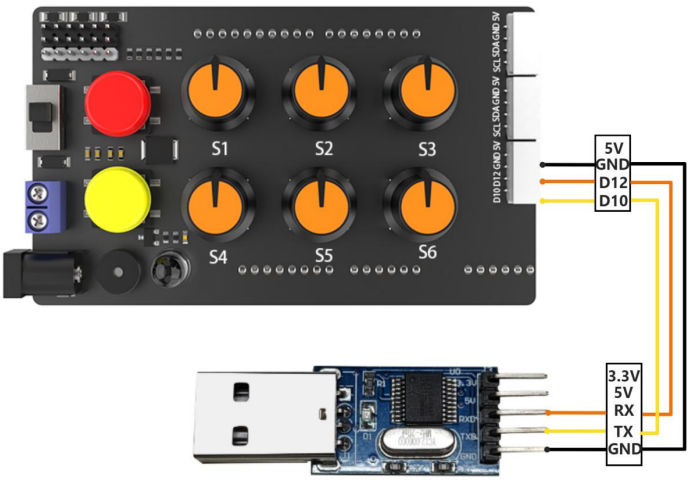
Parity: (Function Code + Data Length + Data) negated, and the low byte is taken.

2. Getting Ready

2.1 Hardware

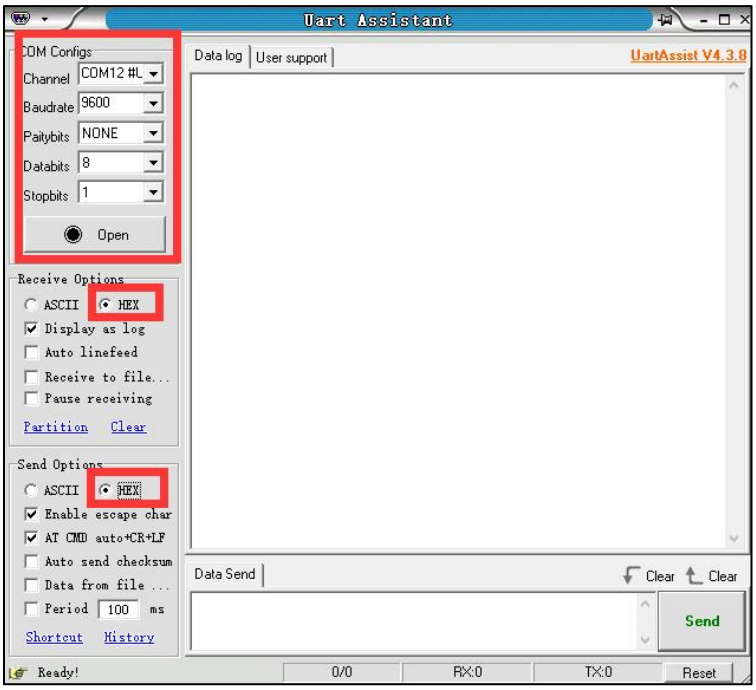
Make sure that the miniArm is properly connected to the PC. Otherwise, it will not communicate properly.

1) Use DuPont wires to respectively connect the RXD, TXD, and GND of the USB adapter to the IO12, IO10, and GND interfaces on the Arduino UNO. Connect the USB adaptor to the PC.



2.2 Software

Make sure the baud is set to 9600, parity to NONE, data bits to 8 and the stop bits to 1 in the serial port utility. Then tick the “hex” to send. The configuration is shown below:



3. Program Outcome

Control the miniArm by sending hexadecimal commands.

1) Command name: “FUNC_SET_SERVO” command value is 0x01, data length is 5:

Instruction: control servo rotation of the miniArm. The rotation angle is 0 to180 degrees.

Data information: five angles required to be set by servo, a total of five values. Each value corresponds to a servo, with a byte to indicate the required rotation angle.

Take sending the hexadecimal sequence “AA 66 01 05 49 16 A1 39 5A 66” as an example. In this sequence, the data information is “49 16 A1 39 5A”, where 49 means the servo NO.1 rotating to 73 degrees, 16 means the servo NO.2 rotating to 22 degrees, A1 means the servo NO.3 rotating to 161 degrees, 39 means the servo NO.4 rotating to 57 degrees, and 5A means the servo NO.5 rotating to 90 degrees.

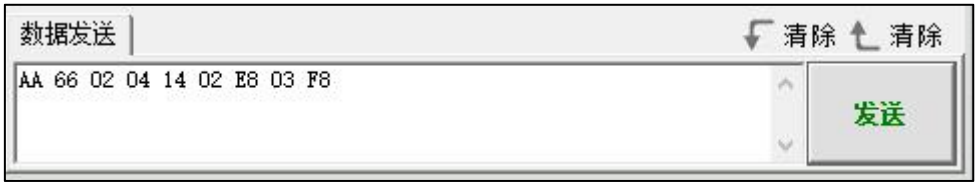
2) Command name: “FUNC_SET_BUZZER” command value is 0x02, data length is 4:

Instruction: control miniArm’s buzzer, frequency, time range from 0 to 65535.

Data information: first is the buzzer frequency, followed by running time, a total of 4 values. Each value is broken down into 2 bytes, with the low byte first and high byte second.

For example, let’s take hexadecimal sequence “AA 66 02 04 14 02 E8

03 F8”. In this sequence, the data information is “14 02 E8 03”. Here the 14 02 represents setting the buzzer frequency to 532, and the E8 03 means the running time is 1000ms.



3) Command name: “FUNC_SET_RGB” command value is 0x03, data length is 3:

Instruction: control the color of the RGB light, where the value range is from 0 to 255.

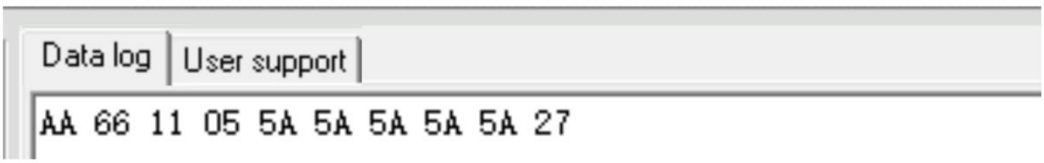
Data information: there are three parameters in total, representing the R(red), G(green), B(blue) colors in RGB respectively.

For example, let’s take hexadecimal sequence “AA 66 03 03 FF 00 00 FA”. In this sequence, the data information is “FF 00 00”. Here, FF represents setting the red color to 255, while the following two 00s means setting the green and blue colors to 0, respectively.

4) Command name: “FUNC_READ_ANGLE” command value is 0x11, data length is 0.

Instruction: read the current miniArm servo angle.

Data information: no data message available since the read operation.

For example, let’s take hexadecimal sequence “AA 66 11 00 EE”.

The miniArm returns to the current five servo angles respectively after sending the data. Take received data “AA 66 11 05 5A 5A 5A 5A 5A 27” as an example. The data message is “5A 5A 5A 5A 5A”, where the 1st to 5th 5A each means the current angle of the 1st to 5th servo is 90 degrees.